

PROJECT OUTPUTS

Data Ingestion:

```
Downloaded to: /root/.cache/kagglehub/datasets/meowmeowmeowmeowmeowmeow/gtsrb-german-traffic-
sign/versions/1
Searching for CSV files in: /root/.cache/kagglehub/datasets/meowmeowmeowmeowmeowmeow/gtsrb-german-traffic-
sign/versions/1
Directory contents: ['test', 'meta', 'Meta.csv', 'train', 'Test', 'Train', 'Meta', 'Train.csv',
'Test.csv']
Found train CSV: /root/.cache/kagglehub/datasets/meowmeowmeowmeowmeowmeow/gtsrb-german-traffic-
sign/versions/1/Train.csv
Found test CSV: /root/.cache/kagglehub/datasets/meowmeowmeowmeowmeowmeow/gtsrb-german-traffic-
sign/versions/1/Test.csv
train: 18839 images copied
test: 6090 images copied
Done! Data ready in model_data/
```

Data:

```
Using device: cuda
Class distribution:
  Class 0: 210 samples
  Class 1: 2220 samples
  Class 2: 2250 samples
  Class 3: 1410 samples
  Class 33: 689 samples
  Class 34: 420 samples
  Class 35: 1200 samples
  Class 36: 390 samples
  Class 37: 210 samples
  Class 38: 2070 samples
  Class 39: 300 samples
  Class 4: 1980 samples
  Class 40: 360 samples
  Class 5: 1860 samples
  Class 6: 420 samples
  Class 7: 1440 samples
  Class 8: 1410 samples

Dataset Summary:
Train samples: 18839
Test samples: 6090
Number of classes: 17
Classes: ['0', '1', '2', '3', '33', '34', '35', '36', '37', '38', '39', '4', '40', '5', '6', '7', '8']
Batch size: 64
Using balanced sampling: Yes
```

Sample images from dataset:
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:627: UserWarning: This DataLoader will create 4 worker processes in total. Our suggested max number of worker in current system is 2, which is smaller than what this DataLoader is going to create. Please be aware that excessive worker creation might get DataLoader running slow or even freeze, lower the worker number to avoid potential slowness/freeze if necessary.
warnings.warn(



Training:

```
Enhanced model setup complete!
Model parameters: 11,336,017

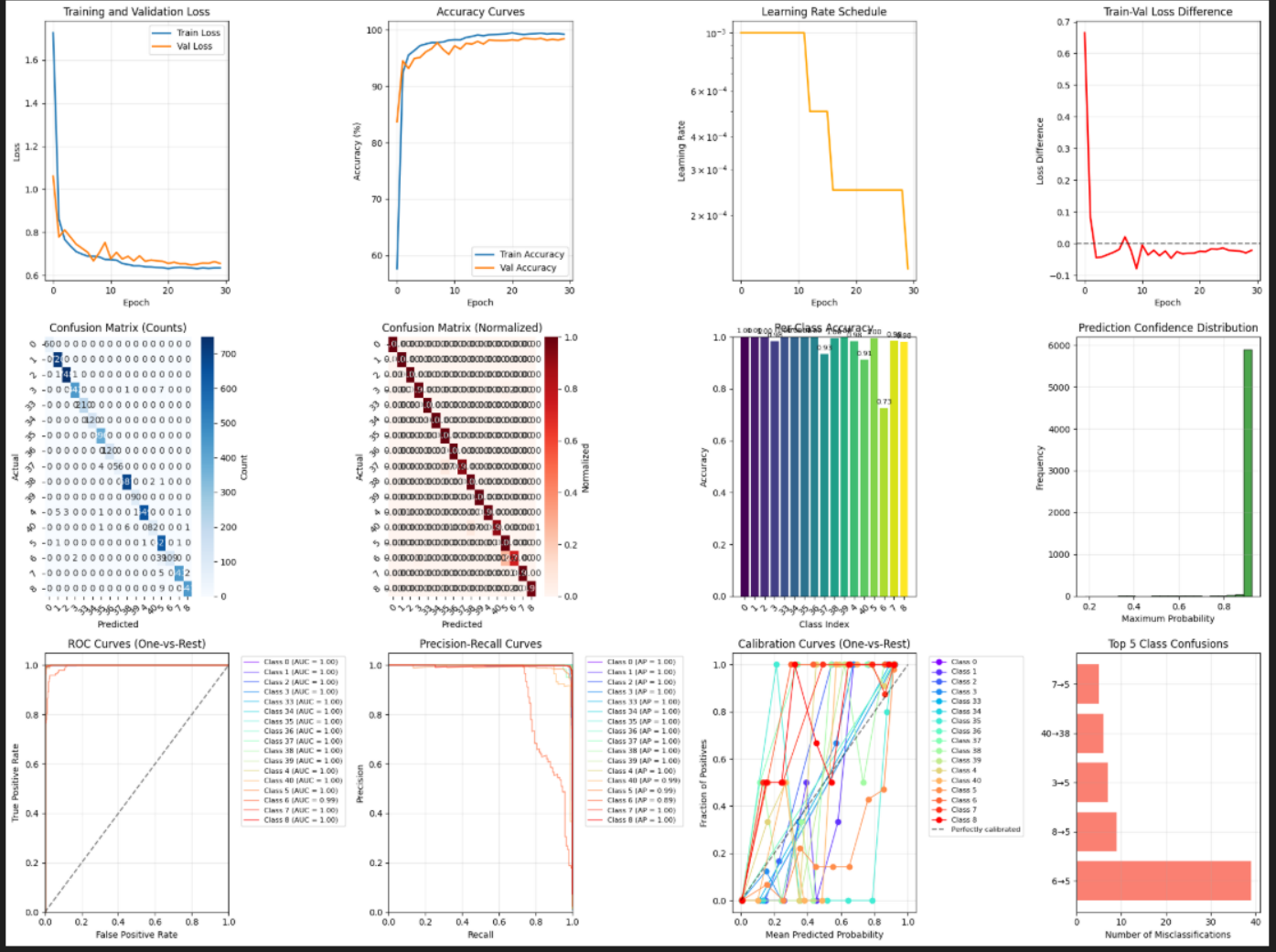
Starting enhanced training...
Training for 30 epochs with early stopping patience: 7
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Epoch 1/30 [Train]: 0% | 0/295 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:62
UserWarning: This DataLoader will create 4 worker processes in total. Our suggested max number of worker in current system is 2, which
is smaller than what this DataLoader is going to create. Please be aware that excessive worker creation might get DataLoader running s
lower, freeze, lower the worker number to avoid potential slowness/freeze if necessary.
  warnings.warn(
Epoch [ 1/30] | Train Loss: 1.7258 | Train Acc: 57.67% | Val Loss: 1.0606 | Val Acc: 83.73% | LR: 0.001000 | Time: 36.5s
✓ New best model saved! Val Acc: 83.73%
-----
Epoch [ 2/30] | Train Loss: 0.8608 | Train Acc: 92.35% | Val Loss: 0.7789 | Val Acc: 94.48% | LR: 0.001000 | Time: 34.7s
✓ New best model saved! Val Acc: 94.48%
-----
Epoch [ 3/30] | Train Loss: 0.7668 | Train Acc: 95.53% | Val Loss: 0.8108 | Val Acc: 93.15% | LR: 0.001000 | Time: 34.4s
-----
Epoch [ 4/30] | Train Loss: 0.7364 | Train Acc: 96.33% | Val Loss: 0.7781 | Val Acc: 94.91% | LR: 0.001000 | Time: 34.2s
✓ New best model saved! Val Acc: 94.91%
-----
Epoch [ 5/30] | Train Loss: 0.7105 | Train Acc: 97.19% | Val Loss: 0.7449 | Val Acc: 95.11% | LR: 0.001000 | Time: 34.8s
✓ New best model saved! Val Acc: 95.11%
-----
Epoch [ 6/30] | Train Loss: 0.6990 | Train Acc: 97.48% | Val Loss: 0.7263 | Val Acc: 96.09% | LR: 0.001000 | Time: 34.0s
✓ New best model saved! Val Acc: 96.09%
-----
Epoch [ 7/30] | Train Loss: 0.6893 | Train Acc: 97.74% | Val Loss: 0.7072 | Val Acc: 96.65% | LR: 0.001000 | Time: 38.4s
✓ New best model saved! Val Acc: 96.65%
-----
Epoch [ 8/30] | Train Loss: 0.6890 | Train Acc: 97.73% | Val Loss: 0.6676 | Val Acc: 97.72% | LR: 0.001000 | Time: 44.3s
✓ New best model saved! Val Acc: 97.72%
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Epoch [ 9/30] | Train Loss: 0.6859 | Train Acc: 97.84% | Val Loss: 0.7049 | Val Acc: 96.52% | LR: 0.001000 | Time: 36.4s
-----
Epoch [10/30] | Train Loss: 0.6742 | Train Acc: 98.16% | Val Loss: 0.7528 | Val Acc: 95.67% | LR: 0.001000 | Time: 37.4s
-----
Epoch [11/30] | Train Loss: 0.6734 | Train Acc: 98.24% | Val Loss: 0.6779 | Val Acc: 97.16% | LR: 0.001000 | Time: 38.1s
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Epoch [12/30] | Train Loss: 0.6711 | Train Acc: 98.22% | Val Loss: 0.7068 | Val Acc: 96.58% | LR: 0.001000 | Time: 39.4s
-----
Epoch [13/30] | Train Loss: 0.6555 | Train Acc: 98.65% | Val Loss: 0.6764 | Val Acc: 97.60% | LR: 0.000500 | Time: 40.1s
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Epoch [14/30] | Train Loss: 0.6509 | Train Acc: 98.82% | Val Loss: 0.6887 | Val Acc: 97.47% | LR: 0.000500 | Time: 39.4s
-----
Epoch [15/30] | Train Loss: 0.6448 | Train Acc: 99.08% | Val Loss: 0.6678 | Val Acc: 97.95% | LR: 0.000500 | Time: 39.0s
✓ New best model saved! Val Acc: 97.95%
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Epoch [16/30] | Train Loss: 0.6449 | Train Acc: 98.94% | Val Loss: 0.6908 | Val Acc: 97.49% | LR: 0.000500 | Time: 39.9s
-----
Epoch [17/30] | Train Loss: 0.6401 | Train Acc: 99.12% | Val Loss: 0.6655 | Val Acc: 98.21% | LR: 0.000250 | Time: 39.6s
✓ New best model saved! Val Acc: 98.21%
-----
Epoch [18/30] | Train Loss: 0.6395 | Train Acc: 99.17% | Val Loss: 0.6716 | Val Acc: 98.11% | LR: 0.000250 | Time: 39.4s
-----
Epoch [19/30] | Train Loss: 0.6372 | Train Acc: 99.23% | Val Loss: 0.6675 | Val Acc: 98.10% | LR: 0.000250 | Time: 38.4s
-----
Epoch [20/30] | Train Loss: 0.6360 | Train Acc: 99.29% | Val Loss: 0.6657 | Val Acc: 98.10% | LR: 0.000250 | Time: 37.1s
-----
Epoch [21/30] | Train Loss: 0.6320 | Train Acc: 99.48% | Val Loss: 0.6558 | Val Acc: 98.23% | LR: 0.000250 | Time: 36.0s
✓ New best model saved! Val Acc: 98.23%
-----
Epoch [22/30] | Train Loss: 0.6361 | Train Acc: 99.27% | Val Loss: 0.6612 | Val Acc: 98.11% | LR: 0.000250 | Time: 36.7s
-----
Epoch [23/30] | Train Loss: 0.6380 | Train Acc: 99.18% | Val Loss: 0.6541 | Val Acc: 98.49% | LR: 0.000250 | Time: 38.7s
✓ New best model saved! Val Acc: 98.49%
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Epoch [24/30] | Train Loss: 0.6368 | Train Acc: 99.27% | Val Loss: 0.6543 | Val Acc: 98.44% | LR: 0.000250 | Time: 40.4s
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Epoch [25/30] | Train Loss: 0.6347 | Train Acc: 99.31% | Val Loss: 0.6483 | Val Acc: 98.36% | LR: 0.000250 | Time: 35.4s
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Epoch [26/30] | Train Loss: 0.6317 | Train Acc: 99.42% | Val Loss: 0.6523 | Val Acc: 98.49% | LR: 0.000250 | Time: 36.0s
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Epoch [27/30] | Train Loss: 0.6350 | Train Acc: 99.26% | Val Loss: 0.6575 | Val Acc: 98.18% | LR: 0.000250 | Time: 35.9s
-----
Epoch [28/30] | Train Loss: 0.6329 | Train Acc: 99.33% | Val Loss: 0.6564 | Val Acc: 98.31% | LR: 0.000250 | Time: 36.4s
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Epoch [29/30] | Train Loss: 0.6349 | Train Acc: 99.32% | Val Loss: 0.6640 | Val Acc: 98.19% | LR: 0.000250 | Time: 35.1s
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Epoch [30/30] | Train Loss: 0.6350 | Train Acc: 99.23% | Val Loss: 0.6557 | Val Acc: 98.42% | LR: 0.000125 | Time: 36.5s

Early stopping triggered after 30 epochs
Best validation accuracy: 98.49%

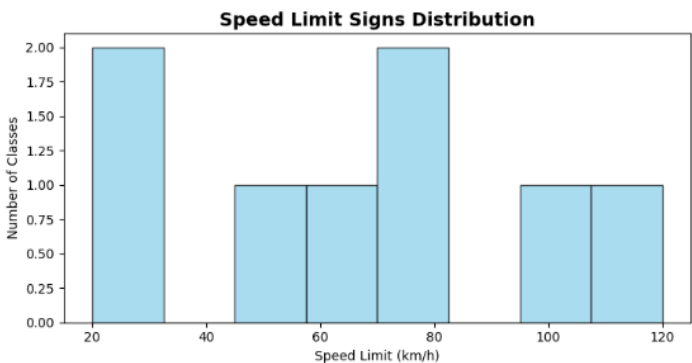
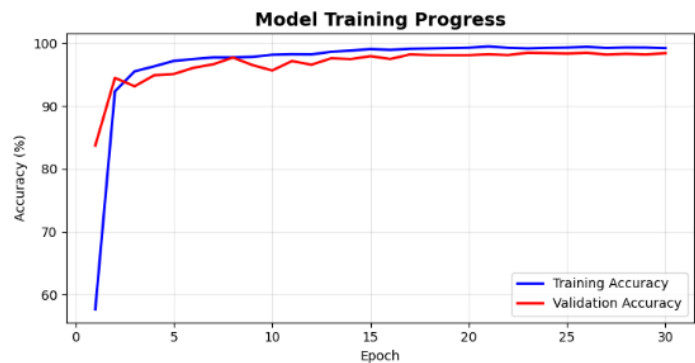
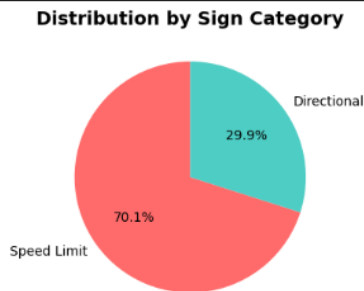
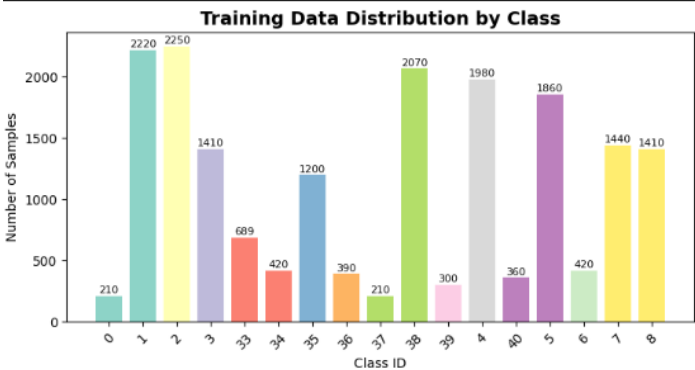
Training completed in 18.8 minutes
```

Evaluation:



CLASSIFICATION REPORT				
	precision	recall	f1-score	support
Class 0	1.00	1.00	1.00	60
Class 1	0.99	1.00	1.00	720
Class 2	1.00	1.00	1.00	750
Class 3	0.99	0.98	0.99	450
Class 33	1.00	1.00	1.00	210
Class 34	1.00	1.00	1.00	120
Class 35	0.98	1.00	0.99	390
Class 36	1.00	1.00	1.00	120
Class 37	1.00	0.93	0.97	60
Class 38	0.99	1.00	0.99	690
Class 39	0.99	1.00	0.99	90
Class 4	1.00	0.98	0.99	660
Class 40	0.98	0.91	0.94	90
Class 5	0.91	1.00	0.95	630
Class 6	1.00	0.73	0.84	150
Class 7	1.00	0.98	0.99	450
Class 8	0.99	0.98	0.99	450
accuracy			0.98	6090
...				
Mean Prediction Confidence: 0.9081 ± 0.0578				

Enhanced model saved as: traffic_sign_model_enhanced_20250919_163423.pth
Metadata saved as: traffic_sign_model_metadata_20250919_163423.json



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🚦  TRAFFIC SIGN CLASSIFICATION MODEL SUMMARY
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📊  MODEL STATISTICS:
• Total Parameters: 11,336,017
• Best Validation Accuracy: 98.49%
• Training Device: cuda
• Total Training Epochs: 30

📁  DATASET INFORMATION:
• Training Samples: 18,839
• Test Samples: 6,090
• Number of Classes: 17

📁  DETAILED CLASS MAPPING:
-----
Index  Class ID  Sign Name                                Category  Samples
-----
0      0      Speed limit (20 km/h)                   Speed Limit  210
1      1      Speed limit (30 km/h)                   Speed Limit  2220
2      2      Speed limit (50 km/h)                   Speed Limit  2250
3      3      Speed limit (60 km/h)                   Speed Limit  1410
4      33     Turn right ahead                        Directional  689
...
• Metadata JSON: traffic_sign_model_metadata_20250919_163423.json
• Loading Code: load_model_code_20250919_163423.py

📁  DOWNLOADING FILES...
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

✅  All files downloaded successfully!

🎉  Enhanced model saving and analysis complete!
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